

20 - Polymerisation

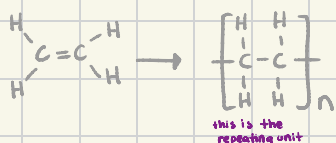
05/01/25

Addition polymerization

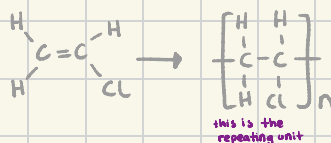
- process where monomers with a double bond (unsaturated) react to form a long-chain polymer w/o loss of any atoms
- the double bond is broken for the polymer to be formed $\star$ for this, a π bond will break

e.g

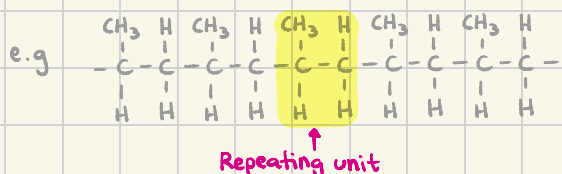
poly(ethene)



poly(chloroethene)
PVC



repeating unit is the part of the polymer that repeats itself, & it's derived from a small part of the monomer



Disposal of poly(alkene) plastics

$\star$ poly(alkene)s are effectively huge alkane molecules which are resistant to chemical attack $\star$

↳ this means that it takes hundred of years for them to decompose; non-biodegradable

↳ a way to solve this problem is burning the plastic BUT this also has disadvantages

- ① complete combustion $\Rightarrow$ releases carbon dioxide which isn't toxic but contributes to global warming
- ② incomplete combustion $\Rightarrow$ releases carbon monoxide which is extremely toxic

$\star$ it's also difficult to separate other plastic waste from poly(alkene)s SO :

$\Rightarrow$ if PVC is burnt, acidic HCl & toxic dioxins are released in the air